Strong and fragile glassformers

Complex disordered systems: Arrested states

The viscosity of a **strong** glassformer follows an Arrhenius law, $\eta = \eta_0 \exp(E_A/k_B T)$, while that of a **fragile** glassformer is often fitted by the Vogel-Fulcher-Tammann (VFT) law

$$\eta(T) = \eta_0 \exp\left(\frac{B}{T - T_0}\right),$$

where B (with units of temperature) and T_0 are empirical parameters.

1. Show that for an Arrhenius law the plot of $\ln \eta$ against $1/T$ is a straight line, and give its slope.
2. For the VFT law, define an *apparent* activation energy through $E_{\text{app}}/k_B = d\ln\eta/d(1/T)$ (the local slope of the same plot). Show that
$$\frac{E_{\text{app}}}{k_B} = \frac{BT^2}{(T - T_0)^2},$$
and describe how E_{app} changes as the liquid is cooled towards T_0.
3. In the Adam-Gibbs picture the relaxation time is $\tau_\alpha = \tau_0 \exp(A/Ts_{\text{conf}})$. Assume that close to the Kauzmann temperature the configurational entropy per particle behaves as $s_{\text{conf}}(T) = K(1 - T_K/T)$, with $K > 0$. Show that the Adam-Gibbs relation then takes exactly the VFT form, and identify B and T_0.

Solution

1. Taking the logarithm
$$\ln \eta = \ln \eta_0 + \frac{E_A}{k_B}\frac{1}{T},$$
which is linear in $1/T$ with slope E_A/k_B: a single barrier controls the dynamics at all temperatures.
2. With $x = 1/T$:
$$\frac{d\ln\eta}{dx} = \frac{d}{dx}\left(\frac{B}{1/x - T_0}\right) = \frac{B}{(1/x - T_0)^2}\frac{1}{x^2} = \frac{BT^2}{(T - T_0)^2}.$$
The slope is no longer constant: the apparent barrier grows on cooling and diverges at T_0. The Angell plot curves upwards, as for fragile glassformers.
3. Substituting s_{conf}:
$$Ts_{\text{conf}} = KT\left(1 - \frac{T_K}{T}\right) = K(T - T_K),$$
so
$$\tau_\alpha = \tau_0 \exp\left(\frac{A}{K(T - T_K)}\right),$$
which is the VFT form with $B = A/K$ and $T_0 = T_K$: the relaxation time diverges where the configurational entropy vanishes. In practice the system falls out of equilibrium at $T_g > T_0$, so the divergence is never reached.